

In this homework we will build on top of the work from Homework 2

1. Use Apache Beam and Cloud Data Flow to do the following data processing on the files in your cloud bucket. First, you should find and print out the top 5 files with the most incoming links, and top 5 files with the most outgoing links (hint: the second should be much easier)
2. Again using Apache Beam and Cloud Data Flow find the top 5 most frequent word bigrams (consecutive word pairs) in the files in your cloud bucket.
3. Run your code both on your local laptop and on the cloud dataflow engine and report on the respective runtimes.

What to turn in:

- A pdf file describing all the necessary steps to configure and run your app, the output of your code and an explanation of what your code does.
- The python code for your app as a github link listed in your report